Quiz

Apr. 27, 2026

多选题 2分

Which of the following statements about priority scheduling and the Completely Fair Scheduler (CFS) are correct?

A. In priority scheduling, a lower integer priority number typically indicates higher priority.

B. Starvation occurs when low-priority processes never execute; aging is a solution that increases priority over time.

C. CFS uses a fixed scheduling quantum for all tasks

D. In CFS, the task with the smallest vruntime value is selected to run next.

E. CFS, an I/O-bound task will have a lower vruntime than a CPU-bound task with the same nice value, giving it higher priority.

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多选题 2分

Which of the following statements correctly compare Round Robin (RR) and Shortest Job First (SJF) scheduling based on the lecture’s comparison table?

A Non-preemptive SJF has a smaller average waiting time than RR.

B RR has a smaller average turnaround time than preemptive SJF.

C RR causes the largest number of context switches among the three algorithms compared.

D Preemptive SJF has a larger average waiting time than non-preemptive SJF.

E Despite worse average waiting and turnaround times, RR provides better responsiveness.

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多选题 2分

Which of the following statements about race conditions are true?

A A race condition occurs when multiple processes/threads access a shared object concurrently.

B The outcome of a race condition depends on the order of accesses to shared resources.

C Race conditions are easy to debug and usually harmless.

D Race conditions can cause incorrect results even if 99% of executions are fine.

E Race conditions only happen on multi-core systems, never on uni-core systems.

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多选题 2分

Which of the following are required properties of a correct critical section implementation?

A Mutual exclusion – No two processes may be inside their critical sections simultaneously.

B Priority inheritance – A low-priority process must inherit the priority of a waiting high-priority process.

C Bounded waiting – There is a limit on how many times other processes can enter their critical sections while a process is waiting.

D Progress – If no process is in its critical section, one of the processes trying to enter will eventually succeed.

E Deadlock freedom – The implementation must avoid all deadlocks.

提交

多选题 2分

Which of the following are true about spin-based locks and sleep-based locks?

A Sleep-based locks are acceptable for short waiting periods, especially on multi-core systems.

B Disabling interrupts is a perfect solution for spin-locks on multi-core systems.

C Sleep-based locks (such as semaphores) typically involve the kernel to put the waiting process to sleep.

D Spin-based locks waste CPU cycles by busy waiting, while sleep-based locks allow the CPU to run other processes.

E Peterson's solution is an example of a sleep-based lock because it uses busy waiting to achieve mutual exclusion.

提交